Note 11.1 Parametrizations of Plane Curves

Parametric Equations

- **Definition**
  If x and y are given as functions

$$x = f(t), y = g(t)$$

  over an interval I of t-values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a parametric curve. The equations are parametric equations for the curve. And the variable t is a parameter for the curve and its domain I is the parameter interval. If I is a closed interval $a \leq t \leq b$, the point $(f(a), g(a))$ is the initial point and $(f(b), g(b))$ is the terminal point. When we give parametric equations and a parameter interval for a curve, we say that we have parametrized the curve. The equations and interval together constitute a parametrization of the curve.

Cycloids

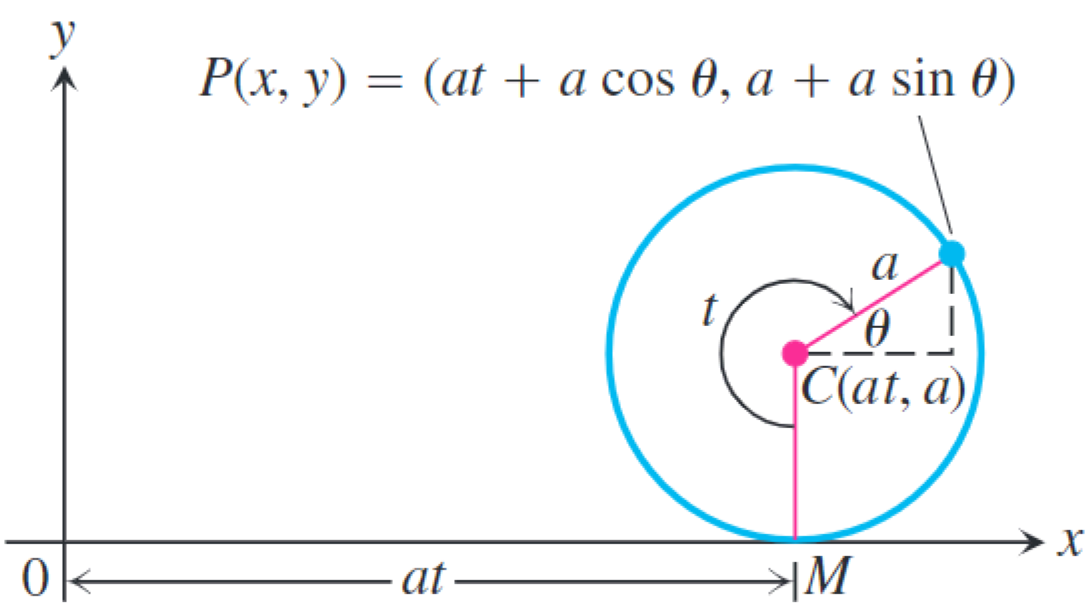

FIGURE 11.8 The position of $P(x, y)$ on the rolling wheel at angle t (Example 8).

A wheel of radius a rolls along a horizontal straight line. Find parametric equations for the path

traced by a point P on the wheel's circumference. The path is called a cycloid.
There the coordinates of P are

$$x = at + a\cos\theta, y = a + a\sin\theta$$

where t is the angle the wheel turns through measured in radians. Then we can express the θ in terms of t, that is

$$\theta = \frac{3\pi}{2} - t$$

Then

$$\cos\theta = -\sin t, \sin\theta = -\cos t$$

Then the parametric equations of the cycloid are

$$x = a(t - \sin t), y = a(1 - \cos t)$$